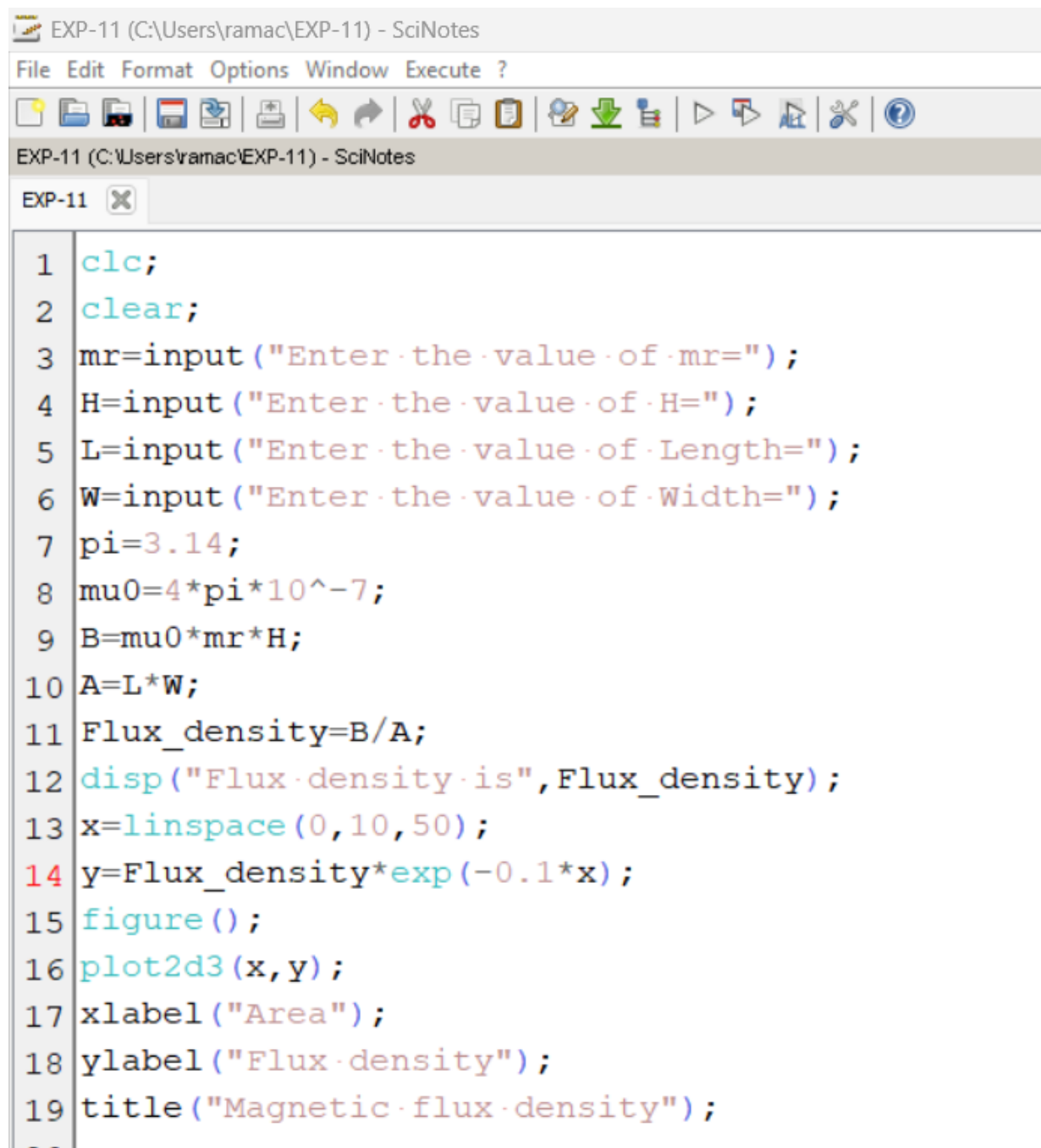


Exp:11 Determination and plotting of Magnetic Flux density in Ferromagnetic Materials with different (i) length, (ii) width and (iii) magnetic field intensity.

Aim: To write program for Magnetic flux density in ferromagnetic materials using SCILAB.

Software required: SCILAB version 6.0.1

Practical output:



```
EXP-11 (C:\Users\ramac\EXP-11) - SciNotes
File Edit Format Options Window Execute ?
EXP-11 (C:\Users\ramac\EXP-11) - SciNotes
EXP-11 [X]
1 clc;
2 clear;
3 mr=input("Enter the value of mr=");
4 H=input("Enter the value of H=");
5 L=input("Enter the value of Length=");
6 W=input("Enter the value of Width=");
7 pi=3.14;
8 mu0=4*pi*10^-7;
9 B=mu0*mr*H;
10 A=L*W;
11 Flux_density=B/A;
12 disp("Flux density is",Flux_density);
13 x=linspace(0,10,50);
14 y=Flux_density*exp(-0.1*x);
15 figure();
16 plot2d3(x,y);
17 xlabel("Area");
18 ylabel("Flux density");
19 title("Magnetic flux density");
```



Scilab 6.1.1 Console

File Edit Control Applications ?



Scilab 6.1.1 Console

Enter the value of $m_r=6$

Enter the value of $H=0.03$

Enter the value of $Length=6 \times 10^{-3}$

Enter the value of $Width=2 \times 10^{-3}$

"Flux density is"

0.0188400

-->



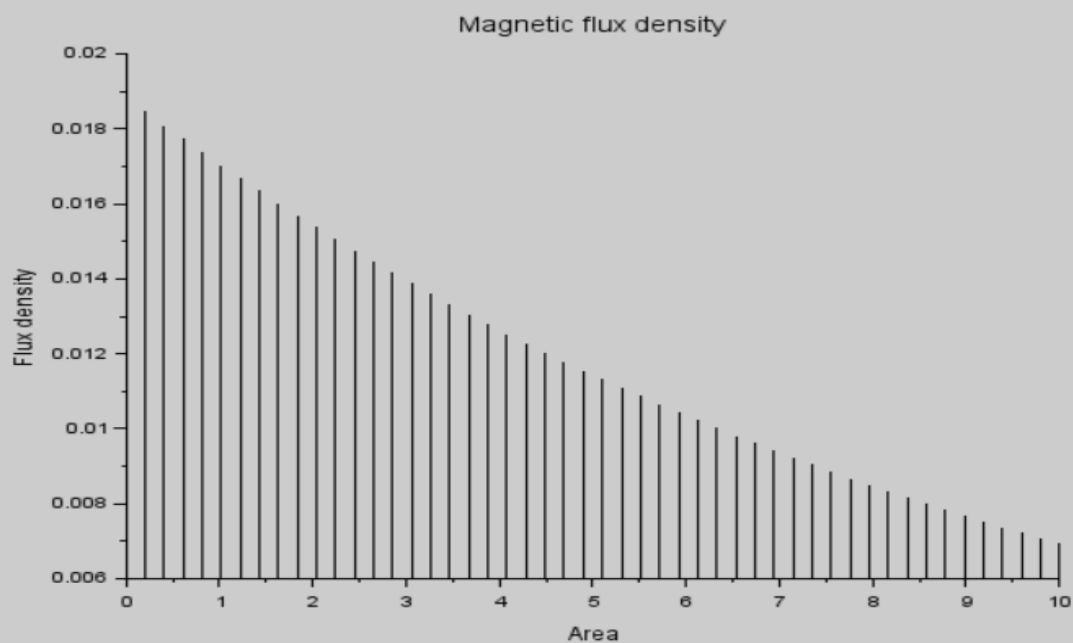
Graphic window number 1



File Tools Edit ?



Graphic window number 1



Theoretical Output:

① $\mu_r = 6$
 $H = 0.03$
length = 6 mm
width = 2 mm

We know that $\mu_0 \Rightarrow 4\pi \times 10^{-7}$

$$B = \mu_0 \mu_r H$$
$$\Rightarrow 4\pi \times 10^{-7} \times 6 \times 0.03$$
$$\Rightarrow 2.2608 \times 10^{-7} \text{ T}$$

Area $\Rightarrow A = l \times w$
 $\Rightarrow (6 \times 10^{-3}) (2 \times 10^{-3})$
 $\Rightarrow 1.2 \times 10^{-5} \text{ m}^2$

flux density $\Rightarrow B/A$
 $\Rightarrow \frac{2.2608 \times 10^{-7}}{1.2 \times 10^{-5}} \Rightarrow 0.01884$
 $\Rightarrow 0.0188$

Result :

Thus, the magnetic flux density was calculated for $\mu_r = 6$, $H = 0.03$ A/m, $L = 6$ mm, and $W = 2$ mm.

The value of Flux Density is found to be 0.01884.